

Unit III: Asymmetric Cryptography

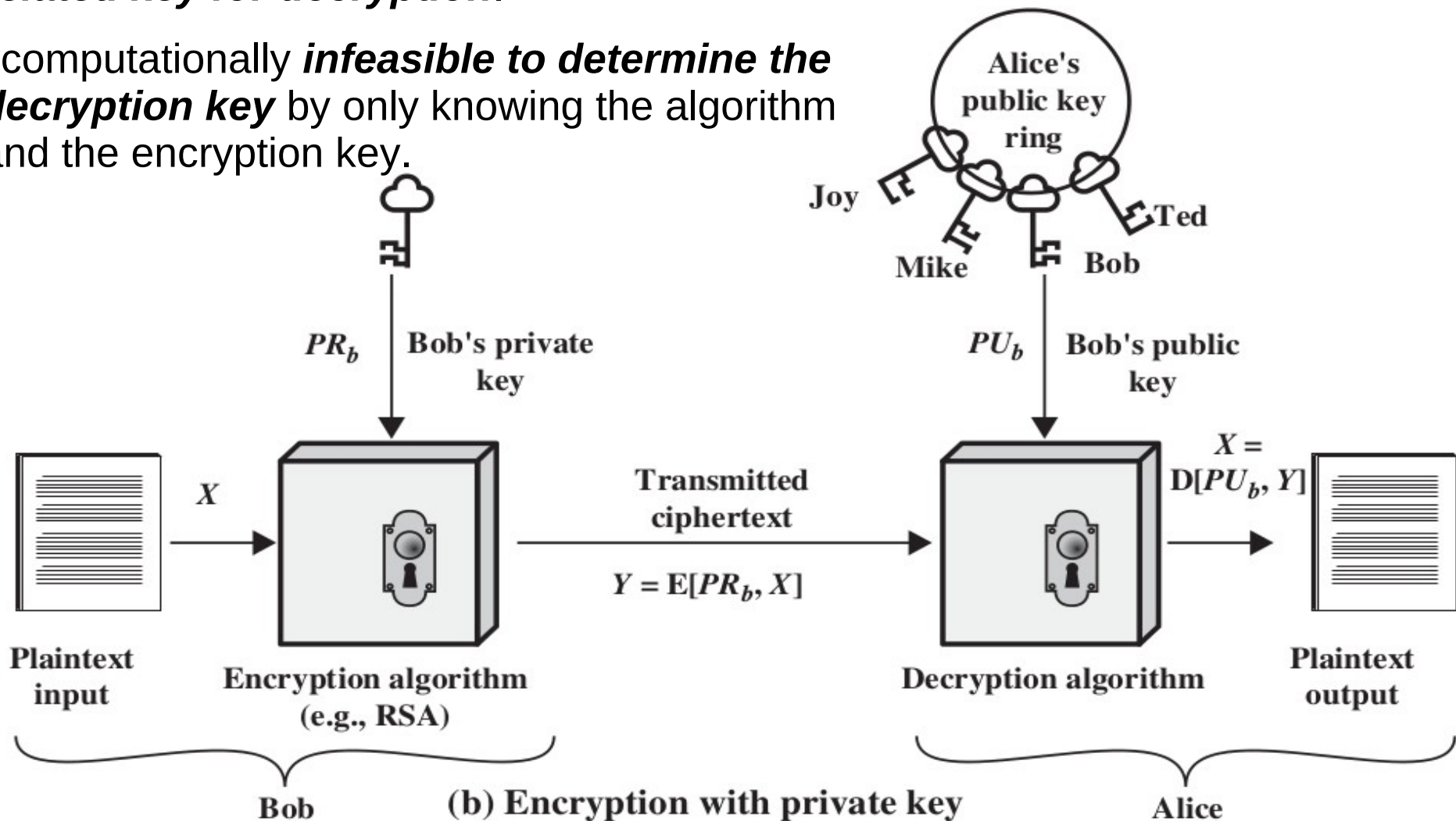
Roshan Chitrakar, PhD

Professor at NCIT
roshanchi@gmail.com

Public-Key Cryptosystems

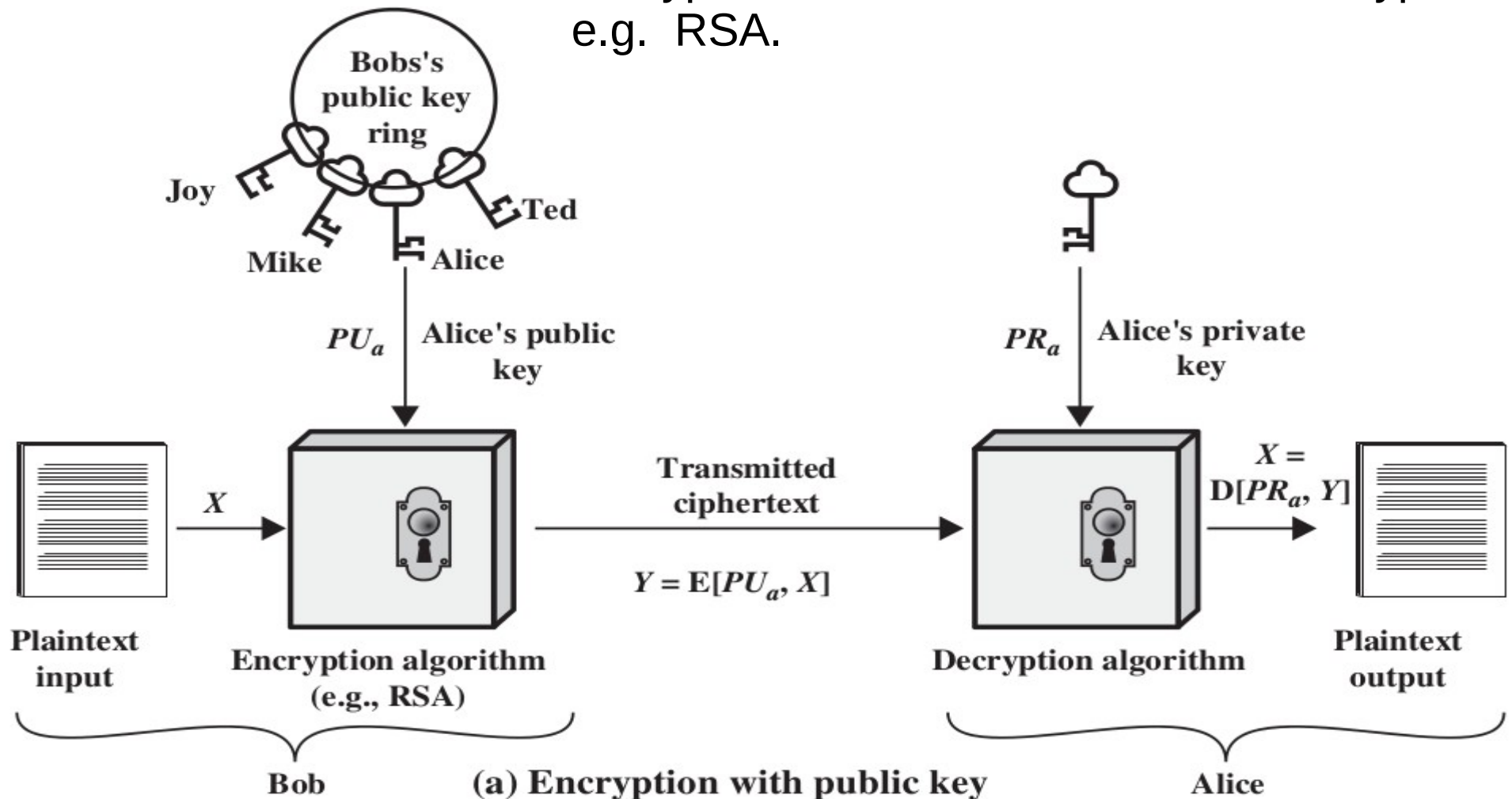
Asymmetric algorithms rely on **one key for encryption** and a **different but related key for decryption**.

It is computationally **infeasible to determine the decryption key** by only knowing the algorithm and the encryption key.



Public-Key Cryptosystems

Either of the two related keys can be used for encryption, with the other used for decryption, e.g. RSA.



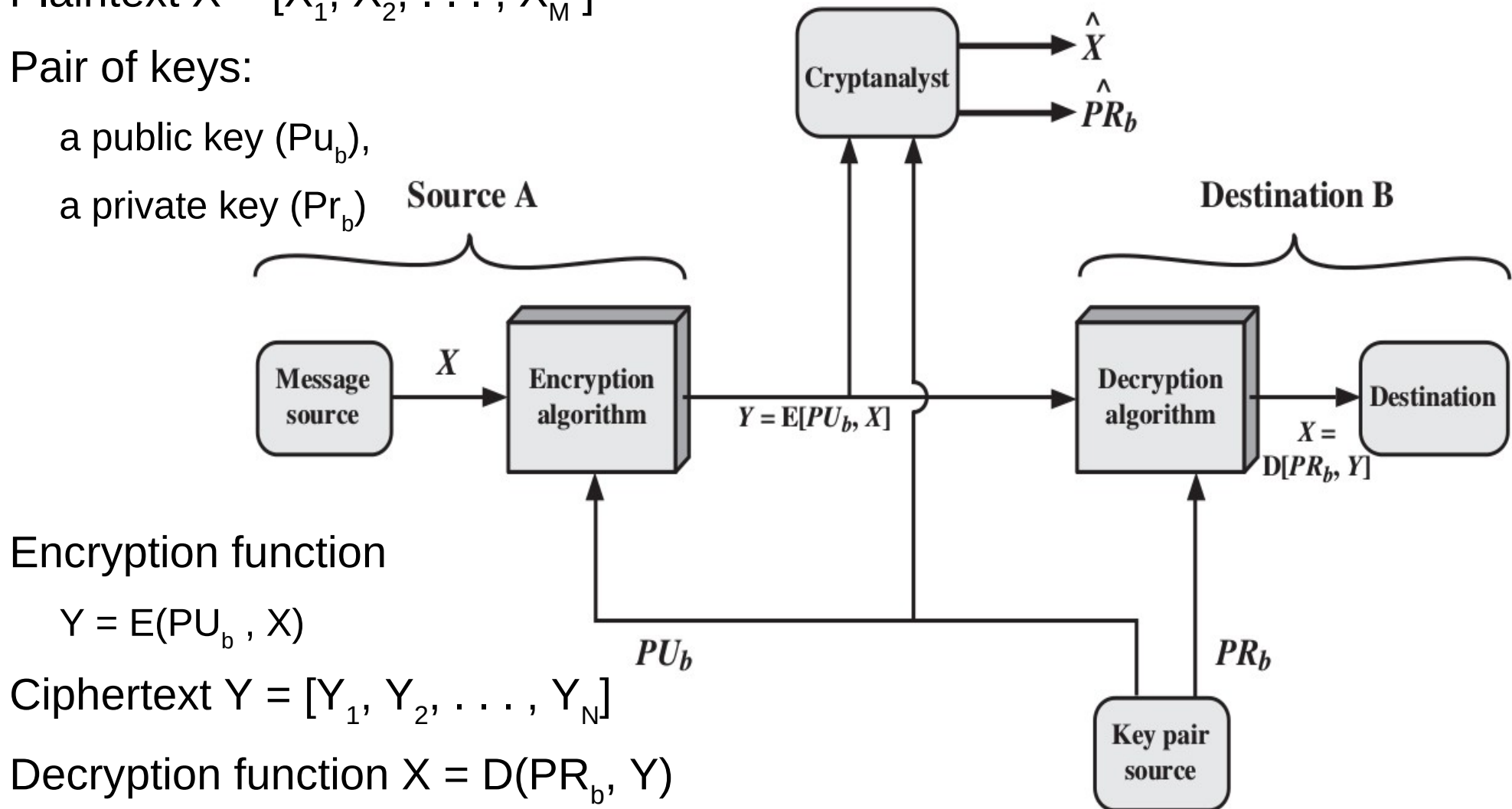
Security Scenario in Public-Key Cryptosystem

Plaintext $X = [X_1, X_2, \dots, X_M]$

Pair of keys:

a public key (PU_b),

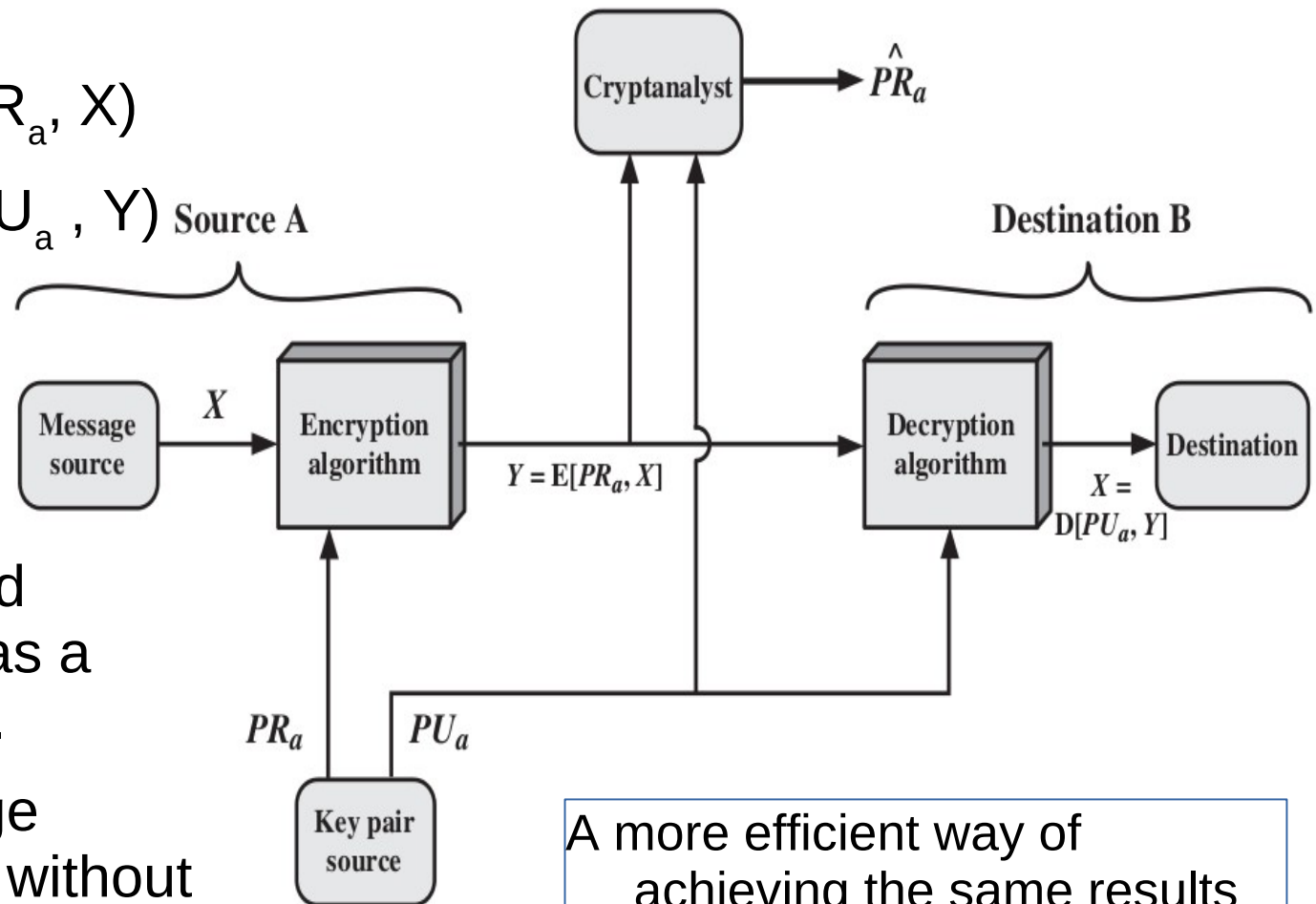
a private key (PR_b)



Authentication with Public-Key Cryptosystem

Encryption $Y = E(PR_a, X)$

Decryption $X = D(PU_a, Y)$ Source A



The entire encrypted message serves as a **digital signature**.

Benefit: the message cannot be altered without access to A's private key and hence ***maintains data integrity***.

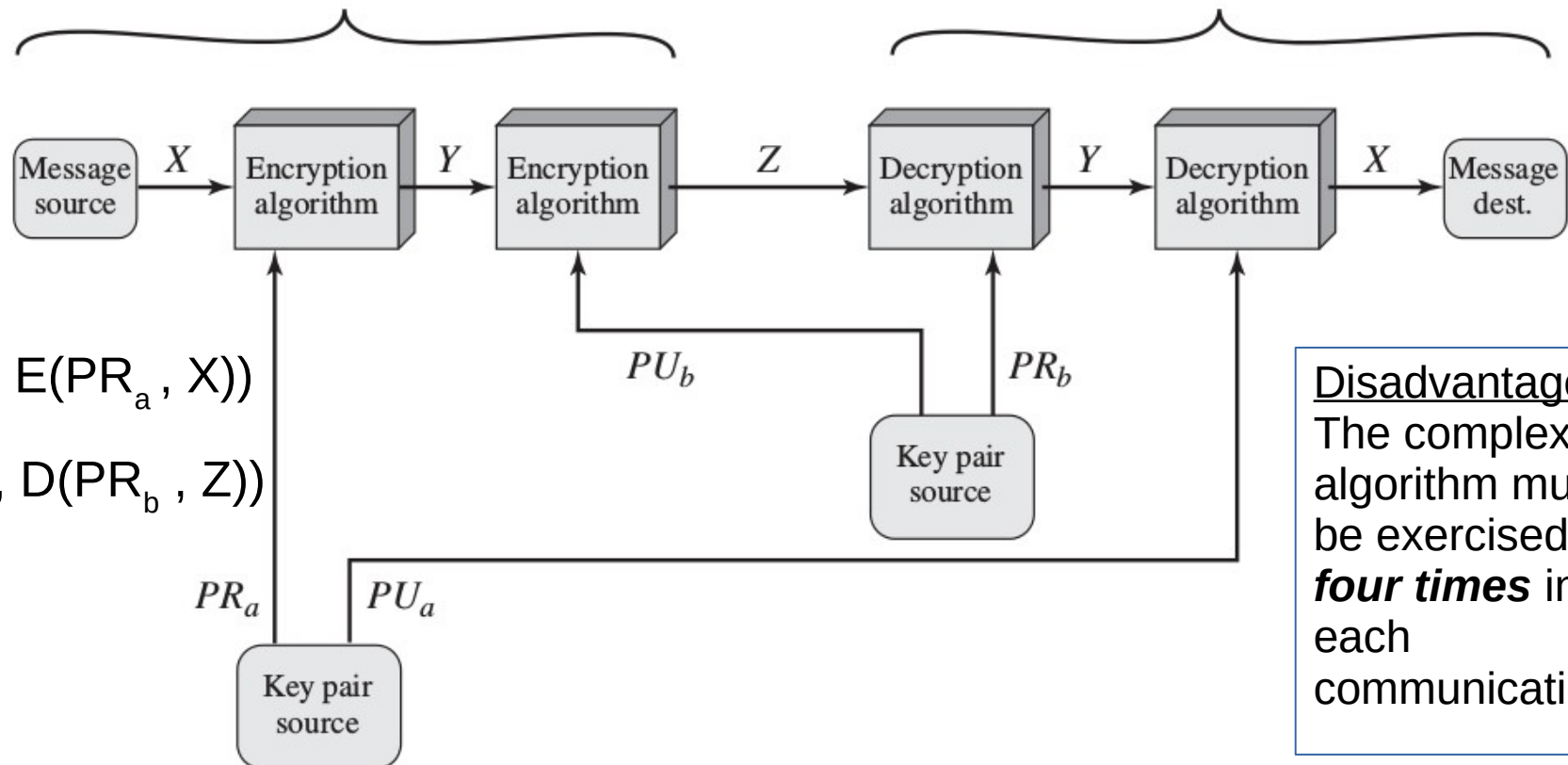
Problem: ***requires more storage***.

A more efficient way of achieving the same results is to encrypt a small block of bits (called ***authenticator***) of the document.

Both Authentication & Confidentiality

Begin by encrypting a message, using the ***sender's private key***, which provides the *digital signature*.

Next, encrypt with the ***receiver's public key***, thus providing the *confidentiality*.



Disadvantage :
The complex algorithm must be exercised ***four times*** in each communication.

RSA Algorithm

Developed in 1977 by ***Ron Rivest, Adi Shamir, and Len Adleman*** at MIT and published in 1978.

The Rivest-Shamir-Adlema (RSA) scheme is a block cipher in which the ***plaintext and ciphertext are integers*** between 0 and $n-1$ for some n .

A typical size for ***n is 1024 bits***, or 309 decimal digits. That is, n is less than 2^{1024} .

It requires ***computation of key pair*** ($K_{\text{pub}}, K_{\text{pr}}$), unlike any symmetric algorithm (AES, 3DES etc.).

Math behind RSA: Euler's Totient Function

- Euler's totient function, written $\phi(n)$, is defined as the number of positive integers less than n and relatively prime to n . By convention, $\phi(1) = 1$.

Example: Determine $\phi(37)$ and $\phi(35)$.

- Since 37 is prime, all of the positive integers from 1 through 36 are relatively prime to 37. Thus $\phi(37) = 36$.
- To determine $\phi(35)$, we list all of the positive integers less than 35 that are relatively prime to it:
1, 2, 3, 4, 6, 8, 9, 11, 12, 13, 16, 17, 18, 19, 22, 23, 24, 26, 27, 29, 31, 32, 33, 34
- There are 24 numbers on the list, so $\phi(35) = 24$.

For a prime number p , $\phi(p) = p - 1$

- Now suppose that we have two prime numbers p and q with $p \neq q$. Then we can show that, for $n = pq$,
- $\phi(n) = \phi(pq) = \phi(p) * \phi(q) = (p - 1) * (q - 1)$
- **For example:** - $\phi(21) = \phi(3) * \phi(7) = (3 - 1) * (7 - 1) = 2 * 6 = 12$
- where the 12 integers are {1, 2, 4, 5, 8, 10, 11, 13, 16, 17, 19, 20}

See also, Euler's
Theorem

RSA : Key Generation

Steps: -

1. Choose **large primes** p, q . (p and $q \geq 2^{512}$)
2. get the **product** $n = p \times q$. ($n \geq 2^{1024}$)
3. And get **prime factorizⁿ** i.e. $\phi(n) = (p-1)(q-1)$.
4. Choose $K_{pub} = e \in \{1, 2, \dots, \phi(n)-1\}$ such that $\gcd(e, \phi(n))=1$.
5. Compute $K_{pr} = d$ such that $d \cdot e \equiv 1 \pmod{\phi(n)}$. (Here, $d \equiv e^{-1}$)
(use Extended Euclidean Algorithm for getting gcd and d)

Here, n may be required for encryption and/or decryption.

Hence, $K_{pub} = (n, e)$ and $K_{pr} = (n, d)$.

RSA : Encryption & Decryption

Encryption

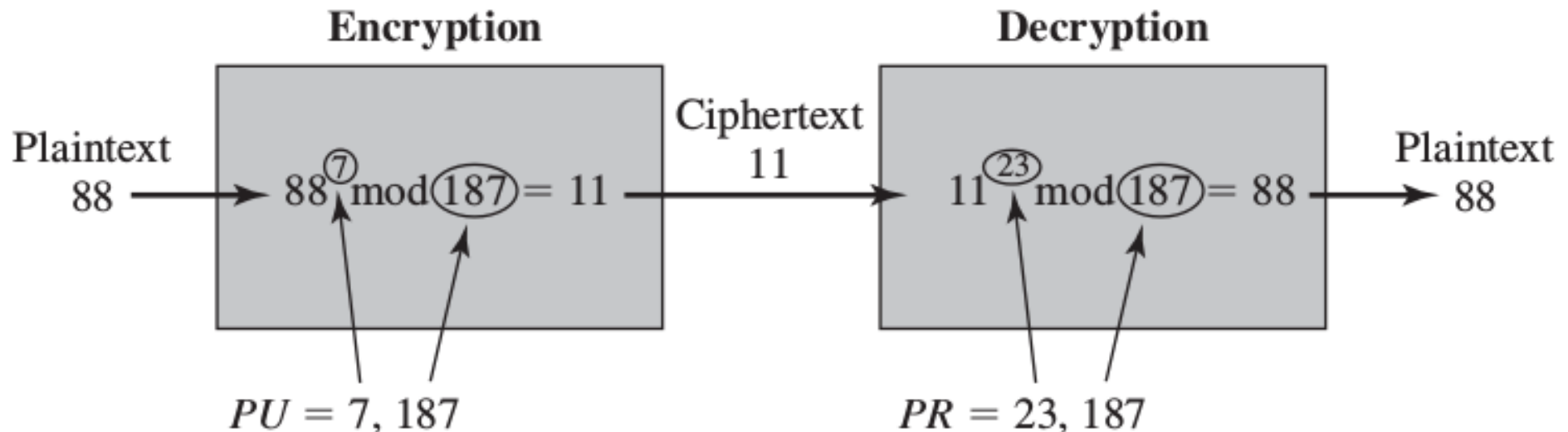
Given $K_{\text{pub}}(n, e)$, $x \in \mathbb{Z}_n = \{0, 1, \dots, n-1\}$,

$$y = e_{K_{\text{pub}}}(x) \equiv x^e \pmod{n}.$$

Decryption

Given $K_{\text{pr}}(d)$, $y \in \mathbb{Z}_n$,

$$x = d_{K_{\text{pr}}}(y) \equiv y^d \pmod{n}.$$



RSA Encryption: Example

Alice

$$x = 4$$

$$x^e = 4^3 \bmod 33 = 31$$

Bob

1. $p = 3, q = 11$

2. $n = 33$

3. $\phi(n) = 2 \times 10 = 20$

4. $e = 3$

5. $d \equiv e^{-1} \equiv 7 \pmod{20}$

$$y = 31$$

$$x = y^d = 31^7 \equiv 4 \bmod 33$$

Efficient Exponentiation

(Applicable to RSA encryption and decryption)

Naive way

$$x^{1024}$$

$$= x \cdot x = x^2$$

$$= x^2 \cdot x = x^3$$

...

...

$$= x^{1023} \cdot x$$

{ involved 1023 multiplications }

Better way

$$x^{1024}$$

$$= x \cdot x = x^2$$

$$= x^2 \cdot x^2 = x^4$$

...

...

$$= x^{512} \cdot x^{512}$$

{ involved 10 multiplications }

Efficient Exponentiation (contd.)

(Square-and-Multiply Algorithm)

Also called **binary method** or **left-to-right** exponentiation.

Rules: -

1. Scan the exponent bit left-to-right
2. In every iteration, do SQUARE.
3. If current bit is 1, MULTIPLY by x.

Example (Decimal):

$$x^{26}$$

$$= (x^1)^2 = x^2$$

$$= x^2 \cdot x = x^3$$

$$= (x^3)^2 = x^6$$

$$= (x^6)^2 = x^{12}$$

$$= x^{12} \cdot x = x^{13}$$

$$= (x^{13})^2 = x^{26}$$

Example (Binary):

$$x^{11010}$$

$$= (x^1)^2 = x^{10}$$

$$= x^{10} \cdot x = x^{11}$$

$$= (x^{11})^2 = x^{110}$$

$$= (x^{110})^2 = x^{1100}$$

$$= x^{1100} \cdot x = x^{1101}$$

$$= (x^{1101})^2 = x^{11010}$$

Diffie-Hellman Key Exchange

Enables two users to ***securely exchange*** a key that can then be used for ***subsequent encryption*** of messages.

The algorithm itself is limited (in practice) to the exchange of secret values only.

The effectiveness of the scheme depends on the ***difficulty of computing discrete logarithms***.

For any integer ***b*** and a primitive root ***a*** of prime number ***p*** , we can find a unique exponent ***i*** such that

$$b \equiv a^i \pmod{p}, \quad \text{where } 0 \leq i \leq (p - 1)$$

The exponent ***i*** is referred to as the discrete logarithm of ***b*** for the base ***a***, mod ***p***, which is expressed as

$$\text{dlog}_{a,p}(b)$$

The Key Exchange Algorithm

Parameters p and α are known to public. Alice and Bob exchanges their keys as follows: -

Alice

$$K_{prA} = a \in \{2, 3, \dots, p-2\}$$

$$K_{pubA} = A \equiv \alpha^a \pmod{p}$$

A is sent to Bob.

$$K_{AB} \equiv B^a \pmod{p}$$

Bob

$$K_{prB} = b \in \{2, 3, \dots, p-2\}$$

$$K_{pubB} = B \equiv \alpha^b \pmod{p}$$

B is sent to Alice.

$$K_{AB} \equiv A^b \pmod{p}$$

Beauty of the DH key exchange is that both communicating parties eventually generate the same key K_{AB} . (*Proof ?*)

The key thus generated may be used as a symmetric key e.g.

$$Y = \text{AES}_{K_{AB}}(x) \quad \text{-----}y\text{-----} \rightarrow \quad \text{AES}_{K_{AB}}^{-1}(y) = x$$

ElGamal Encryption

In 1984, ***T. Elgamal*** announced a public-key scheme based on discrete logarithms, closely related to the Diffie-Hellman technique.

Very simply speaking, ***ElGamal*** cryptosystem is Diffie-Hellman key exchange followed by ***an encryption*** with some ***changes in the order of steps***.

Alice

$$K_{prA} = a \in \{2, 3, \dots, p-2\}$$

$$K_{pubA} = A \equiv \alpha^a \pmod{p}$$

A is sent to Bob.

$$K_{AB} \equiv B^a \pmod{p}$$

Bob

$$K_{prB} = b \in \{2, 3, \dots, p-2\}$$

$$K_{pubB} = B \equiv \alpha^b \pmod{p}$$

B is sent to Alice.

$$K_{AB} \equiv A^b \pmod{p}$$

Alice encrypts plaintext x and sends to Bob, i.e.

$$y \equiv x \cdot K_{AB} \pmod{p} \xrightarrow{y} x \equiv y \cdot K_{AB}^{-1} \pmod{p}$$

{DHKE recalled here}

The ElGamal cryptosystem is ***used in some form in a number of standards*** including the digital signature standard (DSS), and the S/MIME e-mail standard.

ElGamal Encryption (contd.)

Alice

1. Chooses $i \in \{2, 3, \dots, p-2\}$
5. $K_E = \alpha^i \bmod p$ [*where E stands for "Ephemeral Key"*]
6. $K_M = \beta^i \bmod p$ [*M for "Masking Key"*]
7. $y \equiv x \cdot K_M \bmod p$
8. Sends (y, K_E) to Bob

Bob

1. Chooses p , prime element α .
2. $K_{pr} = d \in \{2, 3, \dots, p-2\}$
3. $K_{pub} = \beta = \alpha^d \bmod p$
4. Sends (p, α, β) to Alice.
9. $K_M = K_E^d \bmod p$
10. $x \equiv y \cdot K_M^{-1} \bmod p$

Elliptic Curve CryptoSystem

Why Elliptic Curve?

To find PK family with ***shorter operands***.

Idea: -

To find ***another cyclic group*** in which the DLP is more difficult than the set of prime numbers Z_n ?

Algorithm family	Crypto-system	Security Level (bit)			
		80	128	192	256
Integer factorization	RSA	1024	3072	7680	15360
Discrete Logarithm	DH, ElGamal	1024	3072	7680	15360
Elliptic Curve	ECDH	160	256	384	512
Symmetric key	AES, 3DES	80	128	192	256

Elliptic Curve: Introduction

The equations about circle and ellipse we generally consider are over real numbers $\mathbb{R}$.

To use them for a crypto-system, we need **a polynomial over $\mathbb{Z}_p$** .

The EC over $\mathbb{Z}_p$, $p > 3$, is the set of all pairs $(x, y) \in \mathbb{Z}_p$ and is given by: -

$$y^2 \equiv x^3 + ax + b \pmod{p}$$

Together with **an imaginary point at infinity**, $\mathcal{O}$, where $a, b \in \mathbb{Z}_p$ and

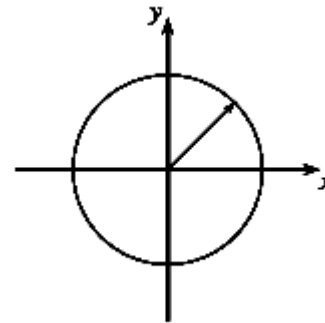
$$4a^3 + 27b^2 \not\equiv 0 \pmod{p}$$

Such an elliptic curve is **symmetric w.r.t. the y-axis**.

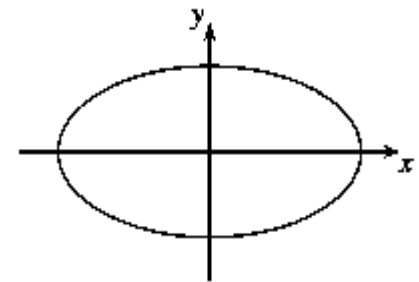
We need to solve the following equation to obtain (x, y) pairs:-

$$y = \pm (x^3 + ax + b)^{\frac{1}{2}}$$

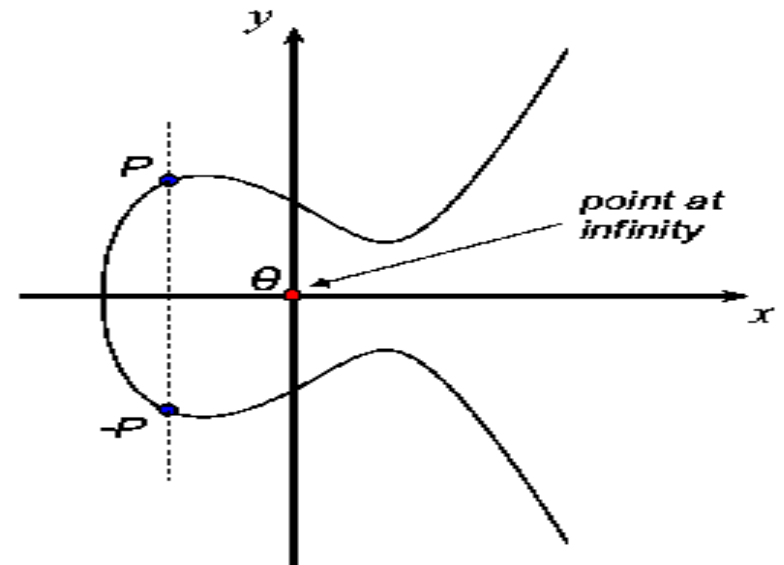
$$x^2 + y^2 = r^2$$



$$ax^2 + by^2 = r^2$$



$$E: y^2 = x^3 - 3x + 3 \text{ over } \mathbb{R}$$



Elliptic Curve DLP

DLP for an EC **requires a cyclic group** which has: -

Set of elements i.e. (x,y) pairs which lie on the curve

Group operations that fulfill group rules.

Group operations: -

Point Addition

- e.g. $P + Q = R$

Point Doubling

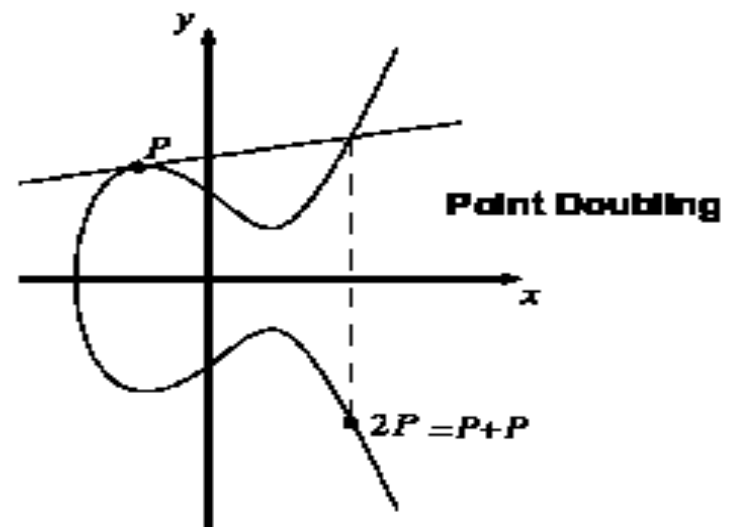
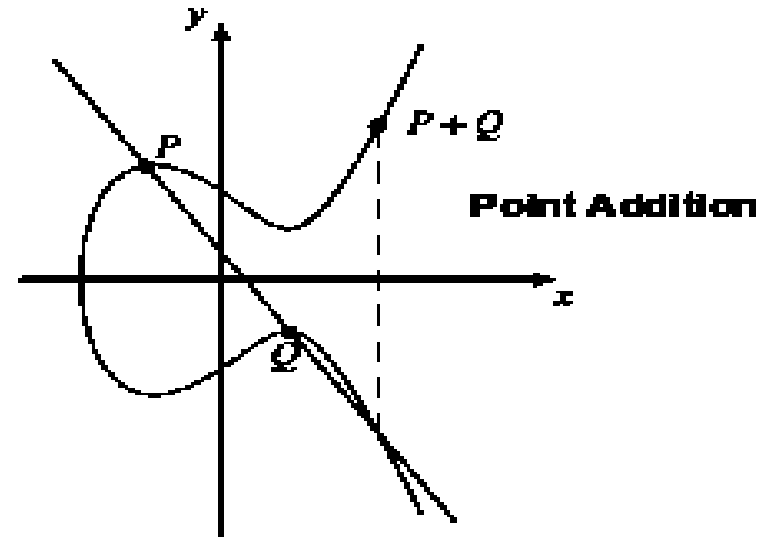
- e.g. $V + V = 2V$

A Neutral element, also called a point at infinity, $\odot$, is defined such that

$$P + \odot = P \text{ for all } P \text{ and}$$

$$P + (-P) = \odot$$

{ Note: here $(-P)$ refers to the point $(x,-y)$.



Elliptic Curve DLP (contd.)

EC as a cyclic group:

The points on an EC, including $\mathcal{O}$, have cyclic subgroups. Under certain conditions, ***all points on an EC form a cyclic group.***

Formal definition of EC DLP:

Given an Elliptic Curve E , we consider a primitive element P and another element T . Now, ***the DLP is to find the integer d*** , where $1 \leq d \leq \#E$, such that

$$P + P + \dots + P = d \cdot P = T$$

Here, d is considered as private key and T as the public key i.e.

$$d = K_{pr} \text{ (an integer value)}$$

$$T = K_{pub} \text{ (a point of the curve i.e. a group element)}$$

Example:

$$E : y^2 = x^3 + 2x + 2 \pmod{17}$$

$$\text{Prime Element } P = (5, 1)$$

Then,

$$2P = P + P = (6, 3)$$

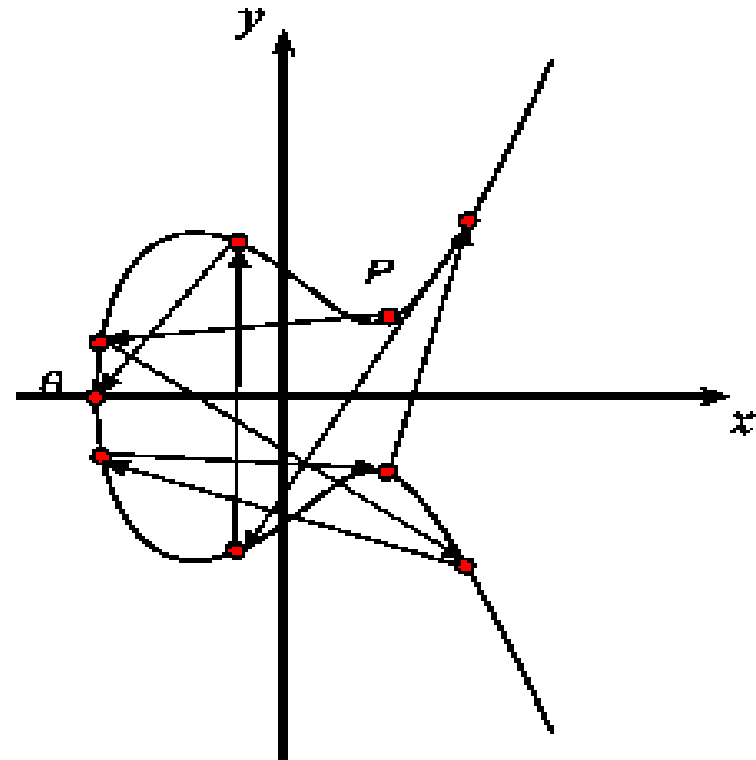
$$3P = 2P + P = (10, 6)$$

...

$$18P = (5, 16) = (5, -1) = -P$$

$$19P = -P + P = \mathcal{O}$$

$$20P = \mathcal{O} + P = P$$



ECDH Key Exchange

An **adoption of DH in Z_p** , which is implemented in two phases: -

Phase I : Setup of parameters viz. curve **$E : y^2 \equiv x^3+ax+b \bmod p$** and the prime element **$P = (x_p, y_p)$**

Phase II : Run the DH protocol

Alice

$$K_{prA} = a \in \{2,3,\dots \dots, \#E-1\}$$

$$K_{pubA} = A \equiv a \cdot P = (x_A, y_A)$$

A is sent to Bob.

$$K_{AB} \equiv a \cdot B = (x_{AB}, y_{AB})$$

Bob

$$K_{prB} = b \in \{2,3,\dots \dots, \#E-1\}$$

$$K_{pubB} = B \equiv b \cdot P = (x_B, y_B)$$

B is sent to Alice.

$$K_{AB} \equiv b \cdot A = (x_{AB}, y_{AB})$$

Note: Point multiplication can be done computationally by using “Double-and-Add” algo.

The co-ordinate thus generated **has a pair of keys** viz. the x-axis key and the y-axis key. **Either one can be used as a symmetric key** e.g.

$$c = \text{AES}_{x_{AB}}(x) \quad \text{-----}c\text{-----} \rightarrow \quad \text{AES}_{x_{AB}}^{-1}(c) = x$$